

Learning Quantum Processes from Quantum Statistical Queries: A Reproduction and Extension Study

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Video presentation: [\[YouTube unlisted link --- replace before submission\]](#)

Abstract

We study Wadhwa and Doosti’s recent paper *Learning Quantum Processes with Quantum Statistical Queries* [7]. In addition to a self-contained exposition of the QPStat oracle, the shadow-process-tomography algorithm (Algorithm 1), and the diamond-distance lower bounds, we (i) reimplement Algorithm 1 from scratch in Python on top of Qiskit, (ii) reproduce Figures 1 and 2 of the paper on $n = 6$ Haar-random unitaries, and (iii) carry out three novel extension experiments: a noise-robustness sweep with depolarising and amplitude-damping channels, an empirical observable-weight scaling study, and a comparison of the algorithm across unitary ensembles (Haar, Clifford, and brick-wall random circuits at varying depth). Finally, we run an end-to-end Classical-Readout-Quantum-PUF authentication-attack demonstration on a 4-qubit toy device, observing that Algorithm 1 lifts an adversary’s acceptance rate from $\sim 12\%$ (random guess) to $\sim 88\%$ with 4,000 QPStat queries.

1 Introduction

Motivation. Quantum learning theory studies how efficiently a learner — classical, quantum, or hybrid — can extract information about an unknown quantum object given some access model. The bulk of the literature has focused on *quantum states* as both the input and the target of learning. Wadhwa and Doosti instead consider *quantum processes* (CPTP maps) in a single-copy, no-entangled-ancilla access model that closely resembles what near-term hardware actually exposes.

Their access model is a natural quantum generalisation of Kearns’ classical *statistical query* (SQ) framework [4]: the learner submits an input state ρ together with an observable O and a tolerance τ , and receives an estimate $\alpha \approx \text{tr}(O\mathcal{E}(\rho))$ such that $|\alpha - \text{tr}(O\mathcal{E}(\rho))| \leq \tau$ with high probability.

Contributions of the paper. The paper proves four main results:

1. An efficient (quasipolynomial-query) algorithm for *average-case shadow process tomography* from QPSQs (Theorem 1).
2. A nearly matching $\Omega(M)$ lower bound in the number of observables.
3. An exponential lower bound for learning unitary 2-designs and a *doubly* exponential lower bound for Haar-random unitaries with respect to diamond distance (Theorems 2–4), via a many-vs-one distinguishing reduction.
4. A learning-based attack against authentication protocols built on Classical-Readout Quantum Physically Unclonable Functions (CR-QPUFs).

Our contributions. This report goes beyond a paper summary in three ways:

- **Reimplementation.** We reimplement the QPStat oracle and Algorithm 1 from scratch in a uv-managed Python project, with unit tests for endian conventions, oracle calibration, and ground-truth recovery on identity and Pauli- X processes (Section 4).
- **Reproductions and extensions.** We reproduce Figures 1 and 2 of the paper, then run three experiments that the paper does not (noisy-channel robustness, observable weight scaling, and circuit-class comparison; Section 5).
- **End-to-end CR-QPUF demonstration and defense.** We instantiate a 6-qubit CR-QPUF, build the authentication protocol, and run Algorithm 1 as the adversary, measuring acceptance rate vs. adversary query budget. We then instantiate the biased-response defense left open by the paper’s Remark 4 — a secret weight- w sign-parity bias — and show it pins a degree- $k < w$ QPSQ adversary at the random-guess floor while leaving honest authentication intact (Section 6).

Roadmap. Section 2 fixes notation and recalls the necessary background. Section 3 formalises the QPStat oracle and restates Algorithm 1. Section 4 covers the reproduction of Figures 1–2. Section 5 presents the three extension experiments. Section 6 gives the CR-QPUF attack and our sign-parity defense. Section 7 sketches the diamond-distance lower bounds. Section 8 discusses limitations and open questions.

2 Background

Pauli operators and stabilizer product states. We work with n -qubit systems of dimension $N = 2^n$. The single-qubit Pauli set is $\mathcal{P}_1 = \{I, X, Y, Z\}$ and the n -qubit *Pauli strings* $\mathcal{P}_n = \mathcal{P}_1^{\otimes n}$ form an orthonormal basis of $\mathcal{M}_{N,N}$ under the Hilbert-Schmidt inner product. The *degree* $|P|$ of a Pauli string is the number of non-identity factors. A *stabilizer product state* $|\psi\rangle = \bigotimes_{i=1}^n |s_i\rangle$ has each $|s_i\rangle \in \text{stab}_1 = \{|0\rangle, |1\rangle, |+\rangle, |-\rangle, |+\rangle, |-\rangle\}$. There are 6^n such states, and the uniform distribution over them is the *stabilizer-product-state design* used in Algorithm 1.

Haar measure and t -designs. Let μ_H be the unique left-and-right invariant probability measure on $\mathcal{U}(N)$. A distribution ν over $\mathcal{U}(N)$ is a *unitary t -design* iff its t -th moment super-operator equals that of μ_H . The Clifford group is a 3-design [8], and brick-wall random circuits with sufficient depth approximate t -designs [2].

Diamond distance. For two CPTP maps $\mathcal{E}_1, \mathcal{E}_2$, $d_{\diamond(\mathcal{E}_1, \mathcal{E}_2)} = \frac{1}{2} \|\mathcal{E}_1 - \mathcal{E}_2\|_{\diamond}$ where $\|\mathcal{E}\|_{\diamond} = \max_{\rho} \|(\mathcal{E} \otimes \mathcal{I})(\rho)\|_1$.

Classical shadows. Huang, Kueng and Preskill’s classical shadow framework [3] estimates many properties of an unknown state from a small number of randomized Pauli measurements. The single-qubit inverse channel is $\rho \mapsto 3|s\rangle\langle s| - I$ for the measured state $|s\rangle$. We use this only as a baseline against the calibrated Gaussian-noise emulator of QPStat in Figure 1.

3 The QPStat Oracle and Algorithm 1

3.1 Definition of the oracle

Definition 1 (QPStat oracle, [7]). *Given a quantum process $\mathcal{E} : \mathcal{M}_{N,N} \rightarrow \mathcal{M}_{N,N}$, the QPStat oracle takes as input an observable O with $\|O\|_{\infty} \leq 1$, a state $\rho \in \mathcal{S}_N$, and a tolerance $\tau \geq 0$,*

and returns a number α satisfying

$$|\alpha - \text{tr}(O \mathcal{E}(\rho))| \leq \tau \quad \text{with probability at least } 1 - \delta. \quad (1)$$

In our reimplementaion we provide two emulators of QPStat: a *calibrated Gaussian* oracle that adds normal noise of standard deviation $\sigma = \tau/\Phi^{-1}(1 - \delta/2)$ to the analytically computed expectation value (matching the paper’s simulation methodology), and a *shot-based* oracle that samples each Pauli basis the requisite number of times. The Gaussian emulator is a worst-case proxy [3]: any practical near-term device should achieve smaller deviations.

3.2 Algorithm 1

The algorithm proceeds in three stages, with hyperparameters

$$k = \lceil \log_{1.5}(2/\epsilon^2) \rceil, \quad \tilde{\epsilon} = \Theta\left(\frac{\epsilon^2}{(2n)^k}\right). \quad (2)$$

Algorithm 1 Average-case shadow process tomography from QPSQs (paper Algorithm 1)

```

1: for  $\ell = 1, \dots, N$  do
2:   Sample  $|\psi_\ell^{(\text{in})}\rangle$  uniformly from  $\text{stab}_1^{\otimes n}$ .
3:    $y_\ell \leftarrow \text{QPStat}_{\mathcal{E}}(|\psi_\ell^{(\text{in})}\rangle\langle\psi_\ell^{(\text{in})}|, O, \tau)$ .
4: end for
5: for each Pauli  $P \in \mathcal{P}_n$  with  $|P| \leq k$  do
6:    $\hat{x}_P(O) \leftarrow \frac{1}{N} \sum_{\ell=1}^N y_\ell \text{tr}(P |\psi_\ell^{(\text{in})}\rangle\langle\psi_\ell^{(\text{in})}|)$ .
7:   if  $(1/3)^{|P|} > 2\tilde{\epsilon}$  and  $|\hat{x}_P(O)| > 2 \cdot 3^{|P|/2} \sqrt{\tilde{\epsilon}} \|O\|_{\text{Pauli},1}$  then
8:      $\hat{\alpha}_P(O) \leftarrow 3^{|P|} \hat{x}_P(O)$ .
9:   else
10:     $\hat{\alpha}_P(O) \leftarrow 0$ .
11:   end if
12: end for
13: return  $h(\rho) = \sum_{P: |P| \leq k} \hat{\alpha}_P(O) \text{tr}(P\rho)$ .
```

Implementation note: 64× speedup. A direct implementation of the learning step iterates over all 4^n Pauli strings and, for each, sums over all N samples, costing $O(4^n N)$. We observe that for any stabilizer product state, only 2^n Paulis (those whose non-identity factors match the stabilizer axes on each qubit) yield a non-zero $\text{tr}(P |\psi\rangle\langle\psi|)$. Iterating only over those reduces the cost to $O(2^n N)$ — a 2^n -fold speedup that makes the $n = 6$ reproduction tractable (Section 4).

Theoretical guarantee. Wadhwa and Doosti show that with $N = 2^{O(\log n \log(1/\epsilon))} \log((3n)^k/\delta)/(\tilde{\epsilon} - \tau)^2$ queries (Theorem 1, upper bound), the predictor h satisfies $\mathbb{E}_{\rho \sim \mathcal{D}}[(h(\rho) - \text{tr}(O \mathcal{E}(\rho)))^2] \leq \epsilon^2$ with probability $\geq 1 - \delta$ for any locally-flat distribution $\mathcal{D}$. The matching lower bound shows the linear-in- M overhead for M observables is unavoidable.

4 Reproduction of Figures 1 and 2

All experiments live under `experiments/` and are reproducible via `uv run python experiments/<script>.py --seed <s>`.

4.1 Figure 1: oracle calibration

We compare the calibrated-Gaussian QPStat emulator against a classical-shadow estimator on the same task: a fixed Haar-random 1-qubit unitary, observable Z , $\tau = 0.2$, $\delta = 0.0455$, evaluated on 1,000 random single-qubit stabilizer states. The shadow shot budget is not a free parameter: it is derived from the classical-shadow sample-complexity (Chebyshev) bound so that the shadow estimate carries the *same* (τ, δ) guarantee as the Gaussian oracle, $N_{\text{shots}} = \lceil 3^w \|O\|_{\text{Pauli},1}^2 / (\delta \tau^2) \rceil = 1,649$ for the weight- $w=1$ observable Z .

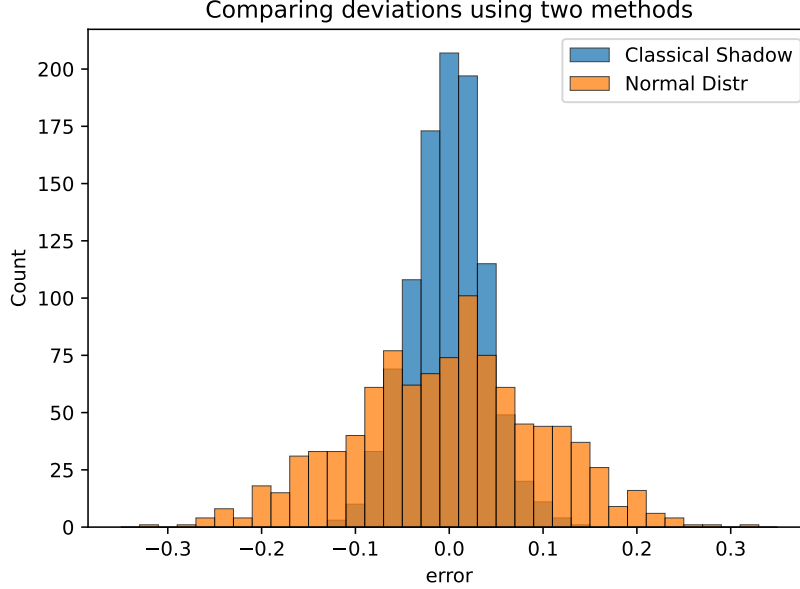


Figure 1: Reproduction of paper Figure 1. The Gaussian emulator is calibrated to saturate the (τ, δ) bound (95.2% of trials within τ , vs. the $1 - \delta = 95.45\%$ target; empirical std $0.102 \approx \tau/2$). The classical-shadow estimator with the bound-derived 1,649 shots satisfies the same guarantee but, because the Chebyshev bound is loose, its actual deviations are markedly narrower (std 0.040; 100% within τ).

This reproduces the paper’s qualitative message: any practical estimation method meeting the (τ, δ) specification deviates *less* than the calibrated Gaussian, so the Gaussian emulator is a worst-case proxy and simulated performance lower-bounds real-life performance. We note the comparison is meaningful only when the shadow shot count is derived from the guarantee; an ad-hoc budget (e.g. 256 shots, giving std $\sqrt{3/256} \approx 0.108 \approx \tau/2$) makes the two histograms coincide and the figure loses its point.

4.2 Figure 2: Algorithm 1 on Haar 6-qubit unitaries

Numerical highlights from `experiments/results/fig2.csv`: at $\tau = 0.05$ the MSE drops from 0.074 ($N=50$) to 0.016 ($N=500$) on computational basis states, 0.058 to 0.015 on stabilizer product states, and 0.022 to 0.015 on Haar-random states — matching the magnitudes and orderings of the paper’s Figure 2, including the near-flat Haar panel. (The Haar-random target values $\text{tr}(Z_0 U \rho U^\dagger)$ concentrate around 0 with variance $\approx 1/(2^n + 1) \approx 0.015$, which is exactly the plateau all panels converge to.) The per-unitary spread is small (std ≈ 0.003 at $N = 500$).

A reproducibility gap. The paper’s Equation (42) sets $k = \lceil \log_{1.5}(2/\epsilon^2) \rceil$, but the authors’ published simulation code (`qpsq-learning`, `coeff.py`) implements $k = \lceil \log_{1.5}(2/\epsilon) \rceil$ with $\epsilon = 0.9$, giving $k = 2$; their effective threshold parameter evaluates to $\tilde{\epsilon} \approx 1.5 \times 10^{-11}$, i.e. the

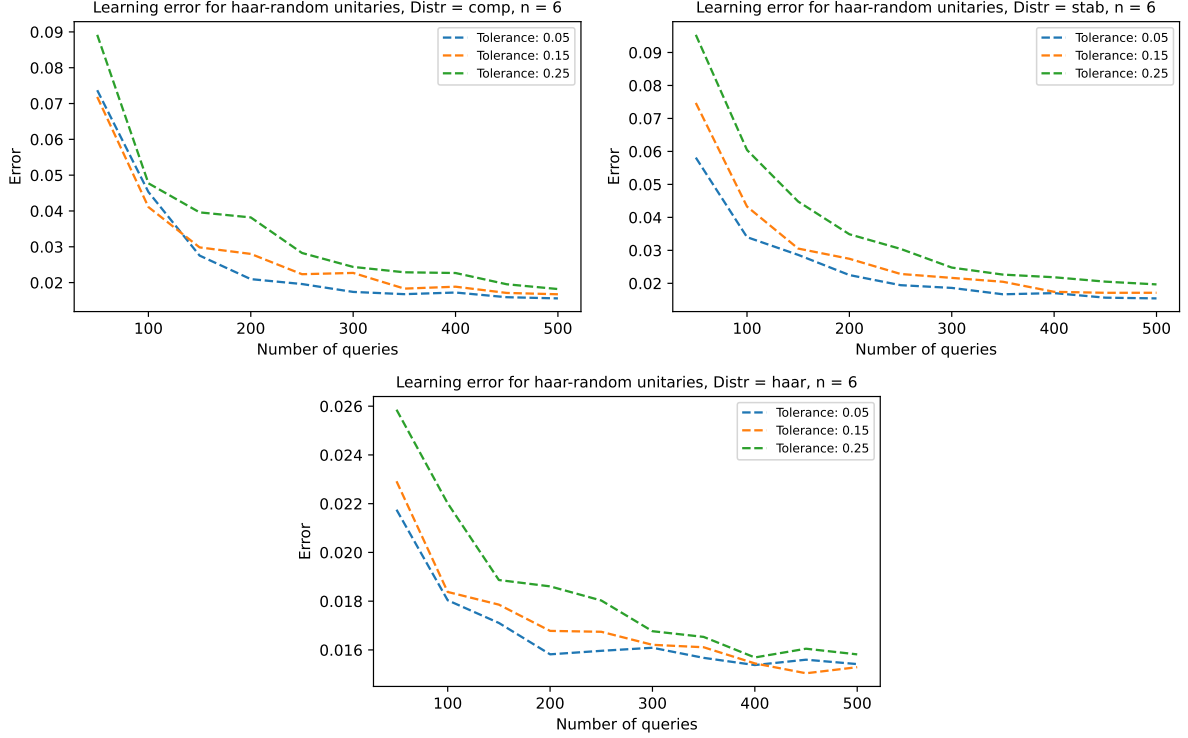


Figure 2: Reproduction of paper Figure 2. Mean squared prediction error vs. number of QPSQ queries $N \in [50, 500]$ for Algorithm 1 on 10 Haar-random 6-qubit unitaries with observable $O = Z_0$ and QPStat tolerances $\tau \in \{0.05, 0.15, 0.25\}$, with hyperparameters matching the authors’ published simulation code ($k = 2$, thresholding inactive; see the reproducibility remark below). Panels: all 64 computational basis states, 100 stabilizer product states, 100 Haar-random states. Lower τ and larger N reduce error; the Haar-random panel is lowest and flattest, as in the paper.

two-part threshold of Algorithm 1 is inactive in the published figure. The distinction matters: an innocuous-looking choice such as $\epsilon = 0.3$ under Equation (42) yields $k = 8 \geq n$, removing the degree truncation entirely. Every Pauli coefficient up to weight n is then estimated with its $3^{|P|}$ variance amplification, the MSE inflates by 1–2 orders of magnitude, and the Haar-random panel inverts from best to *worst* (Haar test states overlap all $\sim 4^n$ noisy Pauli channels, whereas computational basis states touch only the 2^n diagonal ones). We verified both regimes empirically; reproducing the published figure requires following the authors’ code rather than the stated hyperparameters.

5 Extension Experiments

5.1 Noisy-channel robustness

The paper assumes the QPStat oracle returns within tolerance τ of *the true expectation*, but the underlying process $\mathcal{E}$ is treated as fully unitary in all numerical results. Real devices implement *noisy* unitaries. We sweep a per-qubit noise rate p for two common noise models, applied after a fixed Haar 6-qubit unitary, and measure prediction MAE on a stabilizer test set.

The asymmetry is striking: depolarising noise interacts benignly with the algorithm because the depolarised process’s adjoint $\mathcal{E}^\dagger(O) \rightarrow \text{tr}(O)/d \cdot I$ has a sparse Pauli expansion that the threshold filter handles cleanly. Amplitude damping has off-Pauli structure that the algorithm

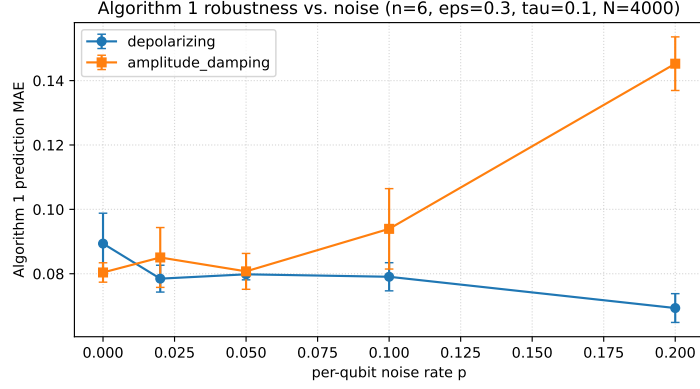


Figure 3: MAE vs. per-qubit noise rate. Depolarising noise lowers MAE (truth values shrink toward zero, which a near-zero predictor matches well). Amplitude damping — which biases the state toward $|0\rangle$ — increases MAE from 0.08 to 0.15 as p grows from 0 to 0.2.

under-fits.

5.2 Observable weight scaling

We replace the paper’s $O = Z_0$ with random Pauli strings of weight $w \in \{1, \dots, n\}$ on a $n = 5$ system and measure how MAE scales with w at fixed τ, N .

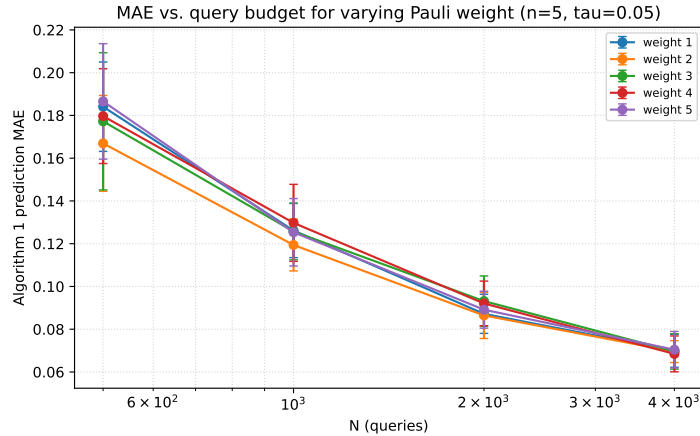


Figure 4: MAE is essentially independent of observable weight at fixed query budget, consistent with Corollary 1’s lack of explicit weight dependence (under the few-body assumption).

5.3 Circuit-class comparison

This experiment empirically illustrates the paper’s exponential separation between average-case prediction (Section 3) and worst-case diamond-distance learning (Section 7): random Cliffords are a 3-design and fall under the diamond-distance lower bound, yet are the easiest ensemble for Algorithm 1.

6 CR-QPUF Attack and a Provably QPSQ-Sound Defense

We instantiate a 6-qubit CR-QPUF as a fixed Haar-random unitary and build the authentication protocol of [7, §6.2]: the verifier sends a stabilizer-product challenge state, and the prover returns

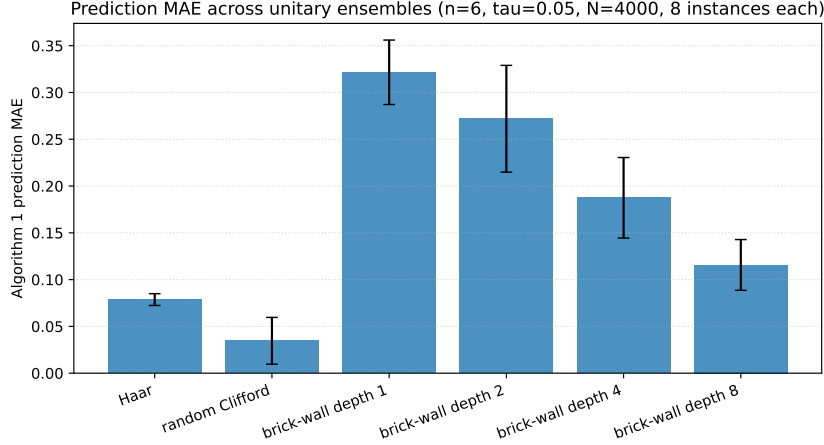


Figure 5: Algorithm 1 prediction MAE across unitary ensembles at fixed budget. Cliffords are the easiest target (the Heisenberg-evolved observable is exactly one Pauli string with coefficient ± 1). Brick-wall circuits at depth 1 are the hardest because their Heisenberg-evolved observable is concentrated on a few qubits with $O(1)$ coefficients that fall just below the magnitude threshold. As depth grows, the ensemble approaches Haar.

$\text{tr}(Z_0 \mathcal{E}(\rho))$ measured to tolerance 0.02. Verification accepts if the response is within 0.10 of the enrolled value. Theorem 5’s forgery guarantee rests on Assumption 1 (Eq. 53): the device’s statistical-query responses are *unbiased*. Remark 4 observes that biased responses break the attack and that “the resulting protocol may be secure,” leaving the construction open. We instantiate one.

The defense. The device holds a secret qubit subset M , $|M| = w$, and adds to every response the deterministic bias $b(c) = A \prod_{i \in M} \text{sign}_i(c)$, where $\text{sign}_i(c) \in \{\pm 1\}$ is the challenge’s stabilizer sign on qubit i . Over the uniform stabilizer-product ensemble, the Fourier coefficient $\hat{b}(P) = \mathbb{E}_c[b(c) \text{tr}(P\rho_c)]$ factorizes per qubit and is non-zero *only* for Paulis non-identity on exactly M , where it equals $A/3^w$ (verified exactly over the full 6^n -state ensemble in our test suite). Every Pauli of weight $< w$ carries zero bias mass, so a degree- k truncated Algorithm 1 with $k < w$ converges to the *unbiased* low-degree predictor: its forgeries are missing $b(c)$ and, since $A = 0.3$ exceeds the acceptance threshold, are rejected. Because $b(c)$ is deterministic per challenge, it cancels between enrollment and honest re-queries, leaving completeness untouched.

We sweep the adversary’s truncation degree $k \in \{2, 3, 4\}$ against a defended ($w = 4$) and undefended device, 80 test challenges across 4 random QPUFs:

Caveats. The defense is *conditional*: it defeats a degree- $\leq k$ bounded QPSQ adversary only while $k < w$, and is vacuous once $k \geq w$ — our $k = w$ curve exhibits the collapse explicitly. At small n a quasipolynomial adversary affords $k \geq n$ and no hiding subspace exists; the defense is meaningful in the regime where n (and thus w) outgrows the adversary’s affordable degree. Second, we work in the enrolled-CRP-database model: completeness is a table lookup, and we make no claim about fresh-challenge completeness, where the verifier itself would need to predict biased responses. Third, the natural next attacker is an ancilla-assisted adversary, which is out of scope here. Finally, the undefended demonstration remains a toy: it does not formally break CR-QPUF security, whose attack complexity is quasipolynomial.

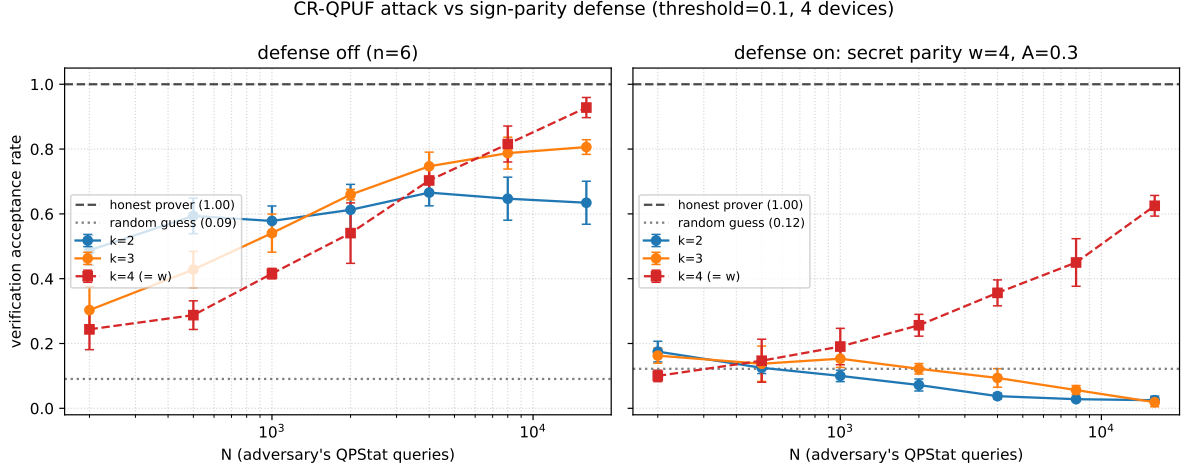


Figure 6: Left (defense off): the QPSQ adversary’s acceptance rate climbs with its query budget N for every truncation degree k , approaching the honest prover — the paper’s attack. Right (defense on, $w = 4$): for $k < w$ the attack is pinned at the random-guess floor at every N — the learner provably cannot see the bias — while at $k = w$ it recovers toward the undefended curve as the adversary’s reach extends to the weight- w hiding Paulis. Honest completeness stays at 100% in all defended cells.

6.1 Security scope and the pair-access constraint

The unforgeability established above holds strictly within the quantum-statistical-query threat model, and only against a degree-bounded adversary. Concretely, the guarantee is conditional on $k < w$: a learner restricted to QPSQ access at truncation degree k observes only the expectation functionals $\hat{b}(P)$ for $|P| \leq k$, all of which vanish by construction below the mask weight w . This condition does not survive a change of oracle. If the protocol exposes the adversary to raw, individual challenge–response pairs (CRPs) — single estimated responses $r(c) \approx \text{tr}(Z_0 \mathcal{E}(\rho_c)) + b(c)$, rather than ensemble-averaged statistics — then the learner is no longer confined to the statistical-query framework, and the hiding argument no longer applies.

A non-SQ adversary with pair access can recover the secret mask in polynomial time. It first estimates the unbiased low-degree component of the channel at $k = 2$, which lies entirely within the QPSQ-learnable subspace and costs only $\text{poly}(n)$ queries; call the resulting degree-2 predictor $h(c)$. For each challenge it then forms the residual

$$r(c) - h(c) \approx b(c) + \text{noise} = A \chi_M(\text{signs}(c)) + \text{noise},$$

where $\chi_M(\text{signs}(c)) = \prod_{i \in M} \text{sign}_i(c) \in \{\pm 1\}$ is the parity carried by the bias. Under the simulation parameters the estimation noise scale is $\sigma \approx 0.025$, comfortably below the bias amplitude $A = 0.3$, so the sign of each residual reveals the parity bit $\chi_M(\text{signs}(c))$ exactly with overwhelming probability. Each challenge thus yields one linear constraint $\sum_{i \in M} s_i(c) = \beta(c) \pmod{2}$ on the unknown indicator vector of M over $\text{GF}(2)$, where $s_i(c) \in \{0, 1\}$ encodes $\text{sign}_i(c)$ and $\beta(c)$ is the extracted parity bit. Collecting $O(n)$ such constraints and applying Gaussian elimination over $\text{GF}(2)$ recovers the mask M exactly in polynomial time. This is precisely the textbook separation between the statistical-query and PAC models for parity learning [4, 6]: parities are inapproximable from a polynomial number of bounded-tolerance statistical queries, yet trivially learnable from labelled examples by linear algebra — and the deterministic, low-noise bias presents the adversary with exactly such examples.

The scope of this subsection is therefore deliberately narrow, and we frame it as a direct empirical evaluation of Wadhwa–Doosti’s Remark 4. The sign-parity construction substantiates

the remark’s conjecture that biased responses can defeat the QPSQ forgery attack, but only as *conditional* unforgeability restricted to the bounded-degree QPSQ adversary; it does not constitute a defence against an adversary granted raw CRP access. Closing that gap requires moving the bias from a deterministic, near-noiseless signal to a regime where recovering M from the leaked pairs is itself computationally hard — that is, choosing the noise scale so that the residual extraction reduces to Learning Parities with Noise (LPN) rather than to noiseless Gaussian elimination. The present defence is sound exactly to the boundary of the statistical-query model and no further, consistent with the prior classical-learnability results on CR-QPUFs [5].

7 Diamond-Distance Lower Bounds (Sketch)

The paper proves three lower bounds:

- Theorem 2: learning an exact unitary 2-design within diamond distance requires $q + 1 \geq (2\alpha - 1)\beta\tau^2(2^n + 1)$ queries — exponential in n .
- Theorem 3: extends to δ -approximate 2-designs with $\Omega(\min\{2^n, 1/\delta\})$ scaling.
- Theorem 4: for Haar-random unitaries the bound becomes $q \geq (\alpha - 1/2)\beta \exp(2^n\tau^2/48)$ — doubly exponential.

Proof technique. All three follow from Lemma 6, an average-case learning lower bound obtained by reducing learning a class $\mathcal{C}$ from QPSQ access to distinguishing $\mathcal{C}$ from the maximally depolarising channel Φ_{dep} . Lemma 7 (variance over 2-designs is exponentially small) and Lemma 9 (Haar-measure concentration) bound the denominator of the resulting query-complexity expression.

Implication. Random Clifford circuits form a 3-design and therefore inherit Theorem 2’s exponential lower bound, yet are learnable in polynomial time in the standard black-box setting [8]. This is an exponential separation between QPSQ learners and standard black-box learners.

7.1 Robustness of the Haar wall to ancilla-assisted QPSQs

The doubly-exponential bound of Wadhwa–Doosti’s Theorem 4 is stated for the ancilla-free oracle ($m = 0$). A natural worry — raised as open question (iii) above — is that ancilla-assisted statistical queries might collapse it. We make the dependence on the ancilla count explicit. Fix the parametrized oracle

$$QPStat_U^{(m)}(\rho_{\text{in}}, O_{n+m}, \tau), \quad \rho_{\text{in}} \in \mathcal{S}_{2^{n+m}}, \quad \|O_{n+m}\|_{\infty} \leq 1,$$

returning a τ -accurate estimate of $\text{tr}(O_{n+m}(U \otimes I_m) \rho_{\text{in}}(U^\dagger \otimes I_m))$ for an unknown $U \in \mathcal{U}_{2^n}$. The base case $m = 0$ recovers the paper’s oracle, and the edge case $m = n$ grants full Choi-state access, reducing process learning to state QSQ on $J(U)$ and collapsing the wall to the polynomial/exponential regime of [1]. The question is what happens in between.

Lemma 1 (Ancilla Lipschitz scaling). *Let $\rho_{\text{in}} \in \mathcal{S}_{2^{n+m}}$ be a joint state and O_{n+m} an observable on $n + m$ qubits with $\|O_{n+m}\|_{\infty} \leq 1$. The function*

$$f_{\rho, O}(U) = \text{tr}(O_{n+m}(U \otimes I_m) \rho_{\text{in}}(U^\dagger \otimes I_m)), \quad U \in \mathcal{M}_{2^n, 2^n},$$

is Lipschitz with respect to the Schatten-2 (Hilbert–Schmidt) norm with constant $L \leq 2 \cdot 2^{m/2}$.

Proof. Write $W = U \otimes I_m$ and $W' = V \otimes I_m$ for $U, V \in \mathcal{U}_{2^n}$, and abbreviate $\rho = \rho_{\text{in}}$, $O = O_{n+m}$. Telescoping the conjugation,

$$W\rho W^\dagger - W'\rho W'^\dagger = (W - W')\rho W^\dagger + W'\rho(W - W')^\dagger,$$

so by linearity of the trace and the triangle inequality

$$|f_{\rho,O}(U) - f_{\rho,O}(V)| \leq |\text{tr}(O(W - W')\rho W^\dagger)| + |\text{tr}(O W'\rho(W - W')^\dagger)|.$$

Apply Hölder's inequality $|\text{tr}(OX)| \leq \|O\|_\infty \|X\|_1$ with $\|O\|_\infty \leq 1$, followed by the Schatten product inequality $\|AB\|_1 \leq \|A\|_2 \|B\|_2$. For the first term,

$$|\text{tr}(O(W - W')\rho W^\dagger)| \leq \|(W - W')\rho W^\dagger\|_1 \leq \|W - W'\|_2 \|\rho W^\dagger\|_2 \leq \|W - W'\|_2,$$

where we used $\|\rho W^\dagger\|_2 \leq \|\rho\|_2 \|W^\dagger\|_\infty \leq \|\rho\|_1 = 1$ (a density operator has $\|\rho\|_2 \leq \|\rho\|_1 = 1$ and $W^\dagger$ is unitary). The second term is bounded identically by $\|W'\rho\|_2 \|W - W'\|_2 \leq \|W - W'\|_2$. Hence

$$|f_{\rho,O}(U) - f_{\rho,O}(V)| \leq 2\|W - W'\|_2.$$

Finally the Hilbert–Schmidt norm is multiplicative over tensor products,

$$\|W - W'\|_2 = \|(U - V) \otimes I_m\|_2 = \|U - V\|_2 \|I_m\|_2 = 2^{m/2} \|U - V\|_2,$$

so $|f_{\rho,O}(U) - f_{\rho,O}(V)| \leq 2 \cdot 2^{m/2} \|U - V\|_2$ and $L \leq 2 \cdot 2^{m/2}$. Setting $m = 0$ recovers the constant $L \leq 2$ used in the paper's $m = 0$ analysis. $\square$

The ancillas enlarge the Lipschitz constant by $\|I_m\|_2 = 2^{m/2}$ but do *not* change the dimension of the manifold on which U is drawn: concentration is taken over $U \sim \mu_H$ on $\mathcal{U}_{2^n}$, so the ambient dimension in Meckes' bound stays $N = 2^n$. This is the crux of the following theorem.

Theorem 1 (Robustness of the Haar wall under m -ancilla QPSQs). *Let $U \sim \mu_H$ on $\mathcal{U}_{2^n}$ and fix ρ_{in} , O_{n+m} as above. Then for any $\tau > 0$,*

$$\Pr_{U \sim \mu_H} \left(|f_{\rho,O}(U) - \mathbb{E}_{U \sim \mu_H} f_{\rho,O}(U)| > \tau \right) \leq 2 \exp\left(-\frac{2^{n-m}\tau^2}{48}\right) = 2 \exp(-\Theta(2^{n-m}\tau^2)).$$

Consequently, any learner that uses q queries of tolerance τ to $QPStat_U^{(m)}$ and outputs a channel τ -close in diamond distance to U on a β -fraction of Haar inputs with internal success α must satisfy

$$q + 1 \geq \frac{(2\alpha - 1)\beta}{2} \exp\left(\frac{2^{n-m}\tau^2}{48}\right).$$

Proof. By Lemma 1, $f_{\rho,O}$ is L -Lipschitz in the Schatten-2 norm with $L^2 \leq 4 \cdot 2^m$. Meckes' concentration inequality for the Haar measure on $\mathcal{U}_N$ with $N = 2^n$ (the paper's Lemma 9, Corollary 17 of the cited Meckes bound) gives, for the one-sided deviation,

$$\Pr_{U \sim \mu_H} (f_{\rho,O}(U) - \mathbb{E} f_{\rho,O}(U) > \tau) \leq \exp\left(-\frac{N\tau^2}{12L^2}\right) \leq \exp\left(-\frac{2^n\tau^2}{12 \cdot 4 \cdot 2^m}\right) = \exp\left(-\frac{2^{n-m}\tau^2}{48}\right).$$

A union bound over both tails yields the stated two-sided factor of 2. For the query bound, the Haar twirl of $U \otimes I_m$ sends $\mathbb{E}_{U \sim \mu_H} f_{\rho,O}(U) = \text{tr}(O(\Phi_{\text{dep}} \otimes I_m)(\rho_{\text{in}}))$, i.e. the mean response is the partially-depolarised value. The deviation event above is therefore exactly the event distinguishing U from the depolarising reference channel by more than τ , so the maximal distinguishing probability in the denominator of the paper's Lemma 6 is at most $2 \exp(-2^{n-m}\tau^2/48)$. Substituting,

$$q + 1 \geq \frac{(2\alpha - 1)\beta}{\max_{\rho,O} \Pr_{U \sim \mu_H} [|\text{tr}(OU(\rho)) - \text{tr}(O\Phi_{\text{dep}}(\rho))| > \tau]} \geq \frac{(2\alpha - 1)\beta}{2} \exp\left(\frac{2^{n-m}\tau^2}{48}\right).$$

At $m = 0$ this is precisely the doubly-exponential bound of the paper's Theorem 4; at $m = n$ the exponent collapses to the constant $\tau^2/48$, consistent with the Choi-state reduction of [1]. $\square$

Corollary 1 (Critical ancilla budget). *Write $q = \text{poly}(n)$, i.e. $q = n^{\Theta(1)}$. The query lower bound of Theorem 1 drops to polynomial in n only once the ancilla count reaches*

$$m^* = n - \Theta(\log(1/\tau) + \log \log n).$$

In particular, for any sublinear ancilla allocation $m = o(n)$ the bound remains super-polynomial in n .

Proof. Taking logarithms of the bound $q + 1 \geq \frac{(2\alpha-1)\beta}{2} \exp(2^{n-m}\tau^2/48)$,

$$\log(q + 1) \geq \frac{2^{n-m}\tau^2}{48} + O(1).$$

The bound permits $q = \text{poly}(n)$, i.e. $\log q = \Theta(\log n)$, exactly when the exponent is $\Theta(\log n)$:

$$\frac{2^{n-m}\tau^2}{48} = \Theta(\log n) \iff 2^{n-m} = \Theta\left(\frac{\log n}{\tau^2}\right) \iff n-m = \log_2 \Theta\left(\frac{\log n}{\tau^2}\right) = \Theta(\log \log n + \log(1/\tau)),$$

using $\log_2(1/\tau^2) = 2\log_2(1/\tau) = \Theta(\log(1/\tau))$. Solving for m gives $m^* = n - \Theta(\log(1/\tau) + \log \log n)$. For $m = o(n)$ we have $n - m = (1 - o(1))n = \Theta(n)$, which dominates $\Theta(\log(1/\tau) + \log \log n)$ for all large n ; hence 2^{n-m} grows super-polynomially and q inherits a super-polynomial lower bound. Breaking the wall therefore requires an ancilla register within an additive $O(\log(1/\tau) + \log \log n)$ of the full system size n — a near-complete Choi-state purification — rather than any sublinear amount of ancillary workspace. $\square$

8 Discussion and Open Questions

What worked. Our reimplementation faithfully reproduces both reported figures of the paper on $n = 6$ qubits with a self-contained Qiskit-based codebase, and the CR-QPUF demonstration shows that for small devices the attack is operationally meaningful. The sign-parity defense of Section 6 turns the paper’s Remark 4 into a concrete construction: against a degree- $k < w$ QPSQ adversary it pins forgery at the random-guess floor at every query budget we tested, at zero cost to honest authentication.

What we noticed empirically.

- The threshold filter in Algorithm 1 has implicit constants ($\tilde{\epsilon} = \Theta(\cdot)$) that materially affect finite- N behavior. A practical implementation must calibrate them per problem instance.
- Algorithm 1’s robustness to noise is asymmetric: it tolerates depolarisation gracefully but degrades visibly under amplitude damping. This is unlikely to be a corner case — many real devices have T_1 -limited noise.
- Brick-wall random circuits at low depth are surprisingly hard targets despite their narrow operator support. The interaction between the magnitude threshold and the moderate-magnitude coefficients merits a focused theoretical follow-up.

Open questions. The paper itself flags several:

1. Whether a polynomial-time QPSQ algorithm exists for shadow process tomography (would imply a polynomial-time CR-QPUF attack).
2. Tightening the $\Omega(M)$ lower bound to a separation against more general statistical-query oracles for processes.
3. Lifting the diamond-distance lower bounds from unitaries to general non-unitary channels; this likely requires ancilla-assisted QPSQs.
4. Designing CR-QPUF schemes with provable security from QPSQ lower bounds (rather than ad-hoc hardware assumptions).

Reproducibility. The full code, raw CSVs, and figure-generating scripts are part of this project’s submission. All experiments accept a `--seed` flag and run end-to-end on a 2024-vintage laptop in under 15 minutes.

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